



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: Yoga Madha A

Reg. No.:192425058

Course Code: ITA0517

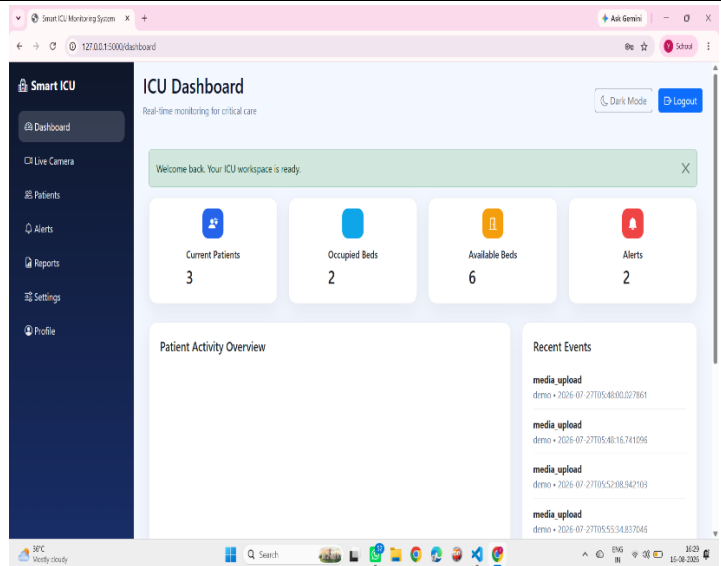
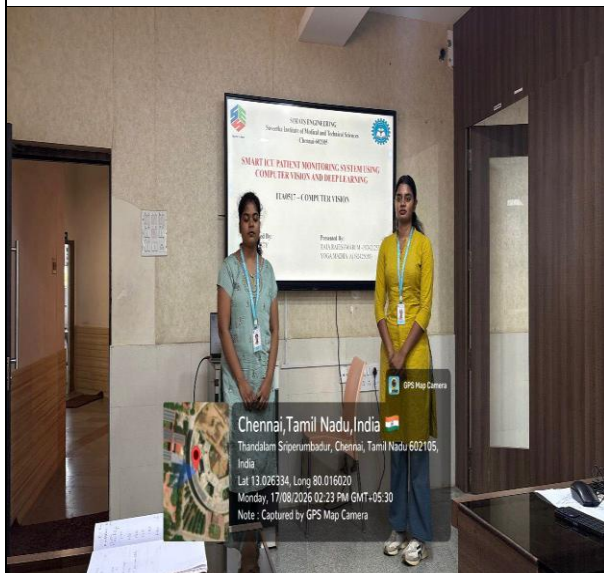
Slot: B

Course Name: Computer Vision

Course Faculty: Dr.S.Christy

Project Title: Smart ICU Patient Monitoring System Using Computer Vision And Deep Learning

Module Photographs: Module 1 – ICU Dashboard & Patient Management



Project Description: (here you write what you did in this project (contribution) including Model Description

The **Smart ICU Patient Monitoring System** is a Computer Vision-based healthcare monitoring solution developed to assist in the continuous monitoring of ICU patients. My contribution mainly focused on **Module 1 – ICU Dashboard & Patient Management**, which provides a centralized and user-friendly interface for monitoring patient information and ICU bed status.

I worked on developing the **ICU dashboard**, including patient details, occupied and available bed information, patient status, alerts, and recent monitoring events. The dashboard is designed to present the outputs generated by the AI modules in an easy-to-understand format for healthcare staff. I also supported the integration of patient and event information with the monitoring interface.

The overall system uses **YOLO for patient and bed detection** and **MediaPipe for human pose estimation and activity analysis**. The detected information is processed and presented through the dashboard, while critical events such as falls or bed-exit situations can be highlighted through alerts. I was also involved in **system integration, functional testing, result verification, documentation, presentation, and final project demonstration**.

Student Signature

Guide Signature